Regis Konan Marcel Djaha

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PROFESSIONAL EXPERIENCE

Research Technician in Statistical Machine Learning, *Basque Center for Applied Mathematics* — 09/2023 – 09/2024, Bilbao, Spain

Mentor: *Iñigo Urteaga, Basque Center for Applied Mathematics,*

- ***Work***: *Deep Latent Variable generative modeling with applications*
- ***Project Description:*** *Development and implementation of Autoencoders (AEs), variational Autoencoders (VAEs), and their multiple extensions to address machine learning problems, including applications in both unsupervised and supervised contexts.*
- ***Objectives:*** *Design and train AE, VAE, and β-VAE models to visualize latent space, generate new data, and perform data compression and reconstruction.*

Research assistant in Mathematical Sciences — 07/2023 – 09/2023, Oxford, United Kingdom

Mfano Africa - Oxford University, Remote

Mentor: *Ramón Nartallo-Kaluarachchi, Oxford University, United Kingdom.*

- ***Project title:*** *Visibility network structure of time-series from real-world systems.*
- ***Project Objectives:*** *Using vertical and horizontal visibility models to develop methods to analyze and interpret complex time-series data from various real-world systems.* Analysis of resting-state fMRI data and financial data using visibility graph methods.

Computational Fluid Dynamics Intern, *Department of Mathematics and Computer Science* — 10/2019 – 02/2020, Abidjan, Ivory Coast

Mentor *: Adou Kablan Jérôme, Université Felix Houphouët-Boigny*

- ***Description:*** *Conducted a comprehensive study on the effects of anisotropic permeability in various mediums on the movement and interaction between freshwater and contaminated water interfaces.*
- ***Objectives***: *Analyze how variations in permeability affect the advancement and stability of the interface between freshwater and dirty water and Numerical study and location of the bevel.*

EDUCATION

Master's Degree in Computational Engineering and Intelligent Systems — 09/2026 – 06/2027, San Sebastián, Spain

University of the Basque Country (UPV/EHU)

Master of Science in Mathematical Science - Machine Intelligence — 02/2023 – 07/2024, Mbour, Senegal

African Institute for Mathematical Sciences (AIMS-AMMI)
Thesis title: Understanding Autoencoders and Variational Autoencoders.
Mentor: Iñigo Urteaga, Basque Center for Applied Mathematics, Spain.
Master's Degree grade: Good Pass

Courses: Mathematics for Machine Learning, Foundations of Machine Learning and Deep Learning, Computer Vision, Natural Language Processing (NLP), Kernel Methods for Machine Learning, and Reinforcement Learning.

Master of Science in Mathematical Sciences — 08/2021 – 06/2022, Kigali, Rwanda

African Institute for Mathematical Sciences (AIMS)
Thesis title: Application of Darcy's law in a 2D porous embankment.
Mentor: Prof Yoshifumi Kimura, Nagoya University, Japan.
Thesis grade: Very Good Pass (Spanish numerical equivalence not available)
Master's Degree grade: Good Pass

Courses: *Advanced Linear Algebra, Classical Mechanics, Operations Research, Partial Differential Equations, Thermodynamics and Atmospheric Physics, Climate Dynamics, Research Methods for Climate, Climate Modeling and climate change, Database and Data Management, Actuarial Mathematics, Numerical methods for Climate Science, Data Assimilation for Climate Science.*

Master's in Mechanics and Energetics, major: Fluid Mechanics — 10/2018 – 02/2020, Abidjan, Ivory Coast

University Felix Houphouët-Boigny (UFHB)
Thesis title: Influence of the anisotropy of the medium's permeability on advancing the freshwater-dirty water interface.
Mentor: Prof ADou Kablan Jerome, University Felix Houphouët-Boigny, Ivory Coast.
Thesis grade: 17/20 .
Master's Degree grade: 13.19/20.

Bachelor's in Mathematics and Applications, major: Mechanics — 01/2015 – 10/2018, Abidjan, Ivory Coast

University Felix Houphouët-Boigny (UFHB)
Bachelor's Degree grade: 10.81/20.

LANGUAGES

- French (native)
- English (B2)
- Spanish (Currently studying)

SKILLS

Programming languages: Scilab, MATLAB, R, C, Python.

Library: Pandas, Numpy, Matplotlib, scikit-learn, PyTorch, TensorFlow.

Software: GNU/Linux, Microsoft Office/Windows, Mac, Server, Git/Github.

Text editors: LaTeX, Word & Libre Office Writer, Markdown.

COURSES

Training in Artificial Intelligence and Data Science ↗ — 10/2024 – 01/2025
Refonte Infini - Infiniment Grand, Remote — London, United Kingdom
Description: *Participation in a customer behavior analysis project for an online retail company as part of a comprehensive training program combining theoretical courses and practical experience.*
Program Focus*: Artificial Intelligence, Generative AI, Model Optimization and Problem-Solving, Statistical Modeling, Rapid Engineering, Exploratory Data Analysis (EDA) and Data Visualization, Machine Learning and Predictive Modeling, Python Data Science, Deep Learning Methods and Techniques.*

Summer School: Machine Learning Crash Course 2024 ↗, — 06/2024
Machine Learning Genoa Center, MaLGa, University of Genova.
Introduction to Statistical Machine Learning, Local Methods and Model Selection, Empirical Risk Minimization with Linear Models, Optimization and SGD, Kernel Methods, Neural Networks, Sparsity and Variable Selection, Dimensionality Reduction, and PCA.

Summer School: Deep Learning and Computer Vision 2024 ↗, — 06/2024
Machine Learning Genoa Center, MaLGa, University of Genova.
Deep Learning Introduction, Image Processing Introduction, Image Features, Motion + Depth, CNN, GANs, Transformers, Group Projects, and Its Applications by using Keras and OpenCV Libraries.

Reproducing Kernel Hilbert Spaces for Machine Learning ↗, *Basque Center for Applied Mathematics (BCAM)* — 10/2023
Supervised learning with linear rules in feature space, Finite dimensional RKHSs. General RKHSs, Supervised learning with RKHSs: uniform concentration, representer theorem, universal consistency, Approximation based on random features, Kernel mean embedding, and maximum mean discrepancy.

PRESENTATIONS

Worshop presentation: Exploring Autoencoders and Variational Autoencoders — 03/2024
Basque Center for Applied Mathematics, Machine Learning Group — Palencia, Spain

Modeling flow in the subsurface — 03/2022
Numerical methods for climate modeling course, African Institute for Mathematical Science — Kigali, Rwanda
Lecturer: Andreas Rupp, Lappeenranta-Lahti University of Technology

The Statistical definition of entropy — 01/2022
Thermodynamics and atmospheric physics course, African Institute for Mathematical Science — Kigali, Rwanda
Lecturer: Bernd Schroers, University of Edinburgh.

AWARDS

International Call IC2023_03 Research Technician in Statistical Machine Learning — 27/09/2023
Funded by La Caixa Junior Leader, €19.188 (EURO)
Basque Center for Applied Mathematics - Bilbao, Spain

Research grant in Mathematical Sciences — 10/07/2023
Mfano Africa-Oxford Virtual Mentorship Programme, £230 (GBP)

Recipient of a full scholarship for the Master's degree — 25/01/2023
Funded by Facebook and Google,
African Master's in Machine Intelligence - Mbour, Senegal

Recipient of a full scholarship for the Master's degree — 30/08/2021
African Institute for Mathematical Science - Kigali, Rwanda, $25.000 (USD)

REFERENCES

Pr. Yoshifumi Kimura, *Lecturer of Fluid Mechanics,* Nagoya University, Japan
kimura@math.nagoya-u.ac.jp, +81 527892819

Dr. Iñigo Urteaga, *Ikerbasque Research Fellow,* Basque Center for Applied Mathematics, Spain
iurteaga@bcamath.org, +34 626 65 29 37

CERTIFICATES

Data Science Workshop: From theory to Practice
African Institute for Mathematical Sciences Research Innovation Centre

Participation in the national phase of the second edition of the CAMES Olympiad
African and Malagasy Council for Higher Education (CAMES), University Felix Houphouët-Boigny (UFHB).

Virtual Mentorship Programme in Mathematical Sciences
Mfano Africa-Oxford Mathematics.

Reproducing Kernel Hilbert Spaces for Machine Learning
Basque Center for Applied Mathematics (BCAM), Spain

Training in Python and Scilab: Application to the study of Partial Differential Equations
Mathematical Society of Ivory Coast (SMCI), Mathematical School of Yamoussoukro (EMY), Ivory Coast.

PUBLICATIONS

Dieu-Donné Fangnon, Armandine Sorel KOUYIM, Verlon Roel MBINGUI, Phanie Dianelle NEGHO, and ***Regis Konan Marcel Djaha.*** A comparative study of Bitcoin and Ripple's cryptocurrencies trading using Deep Reinforcement Learning, Poster and Publication at 6th Deep Learning Indaba Conference 2024 (DLI 2024), Senegal, 01-07 September 2024, https://arxiv.org/pdf/2505.07660 ↗ .

John Kamwele Mutinda, Amos Langat, ***Regis Konan Marcel Djaha****, Jackson Ndoto Munyao, Lee Whitaker, and Millicent Auma Omondi.* Enhancing Obesity Detection Through SMOTE -based Classification Models: A Comparative Study, Interdisciplinary Journal of Epidemiology and Public Health - IJEPH, June 30, 2024, https://revistas.unilibre.edu.co/index.php/iJEPH/article/view/11532 ↗ .